

# Dheeren Tejani

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**Portfolio:** dheerentejani.netlify.app | **Location:** Mumbai, India

## Summary

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AI Engineer focused on Generative AI and efficient ML systems. Experienced in building and optimizing LLMs and diffusion models from scratch, and deploying production-grade inference pipelines, reducing costs near **7x** through model quantization and hardware-aware optimization.

## Experience

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### AI Engineering Intern

*Ally Blocksecure Media Pvt. Ltd. (TheAlly)*

*Jan 2026 – Jun 2026*

- Built an end-to-end PDF parsing pipeline, replacing a 70B LLaMA-based system with a 4-bit quantized 7B Qwen model, achieving **10x parameter reduction** and cutting inference costs by nearly **7x** (2.99 USD/hr on H100 → 0.39 USD/hr on L4). Developed a FastAPI backend with async job queuing and encryption, handling the full pipeline from document preprocessing to structured output postprocessing.

### Full Stack Developer

*Vivekanand Education Society (VESASC)*

*Jul 2025 – March 2026*

- Developed the official **Exam Remuneration System for both written and practical examinations** (Serverless), digitalizing workflows for **100+ faculty**, replacing manual paper processes with a centralized, secure online platform.

## Projects

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### LLaMa style Large Language model

*PyTorch, Transformers, React*

- Built a 124M-parameter decoder-only Transformer from scratch in PyTorch (self-attention, positional embeddings, causal masking). Pretrained on Cosmopedia dataset over **84K steps / 22B tokens at 400K tokens/sec** on a H100 using BF16 AMP, gradient accumulation, cosine LR scheduling, and FlashAttention. Model demonstrated **Instruction-following capability** after SFT, achieving a final loss of **1.3**. Hosted it with serverless FastAPI Backend with rate limiting and React Frontend.

### Diffusion Transformer (DiT) — Image Generation model

*PyTorch, Flow Matching, RoPE*

- Built a **1.6B-parameter** Diffusion Transformer from scratch with flow matching, 2D RoPE positional encodings, Heun sampler, and timestep shift. Pre-latentized **6M Danbooru images** using Flux's 16-channel VAE across 5 aspect ratio buckets. Trained for **14K steps** on a B200 GPU, achieving a final MSE loss of **0.50** (lowest 0.48). Model successfully learned global color distribution, contrast, and style coherence.

## Skills

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**Languages:** Python, JavaScript/TypeScript, SQL

**AI/ML:** PyTorch, Computer Vision, NLP, Model Optimization, Flow Matching, MLOps, Quantization, Transformers, LLMs

**Deployment:** RunPod, FastAPI, Streamlit, Docker, Git, AWS, Modal

**Web Dev:** React, TypeScript, Tailwind CSS

## Education

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### B.Sc. in Artificial Intelligence

*Jun 2023 – Apr 2026*

**College :** Vivekanand Education Society's College of Arts, Science and Commerce (Autonomous) Mumbai

**CGPA :** 8.65